Multi-Modal Document Understanding using Layout-Aware Transformers

Abstract:

Extracting structured information from unstructured documents such as invoices and forms remains a challenging problem due to the combination of textual, visual, and spatial information. In this work, we propose a multi-modal pipeline that integrates OCR and layout-aware transformer models to improve information extraction accuracy. Our approach demonstrates how combining textual and positional features enhances performance in document understanding tasks.

Introduction:

Unstructured documents such as PDFs, scanned forms, and invoices contain valuable information but lack a predefined schema. Traditional NLP approaches fail to capture spatial relationships within documents. Recent advances in transformer-based models, such as LayoutLM, enable the integration of text and layout information, making them suitable for document understanding tasks.

Methodology:

We designed a pipeline consisting of three components: OCR-based text extraction, layout-aware feature encoding, and transformer-based information extraction. OCR was used to extract textual content along with bounding box coordinates. These were fed into a LayoutLM-based model to capture both textual and spatial relationships. The model was fine-tuned to extract key-value pairs from document samples.

Experiments:

A custom dataset was curated by annotating document samples such as invoices and forms. The model was trained and evaluated on this dataset using standard metrics such as precision, recall, and F1-score. Experimental results showed improved extraction accuracy compared to baseline text-only approaches.

Conclusion:

This work highlights the effectiveness of multi-modal approaches in document understanding tasks. By integrating OCR with layout-aware transformers, the system achieves better performance in extracting structured information from complex documents. Future work includes scaling the dataset and exploring more advanced multi-modal architectures.